

# 1. Vehicle Maintenance Scheduler

## Description

A backend scheduling system developed using Express.js for managing vehicle maintenance tasks across depots.

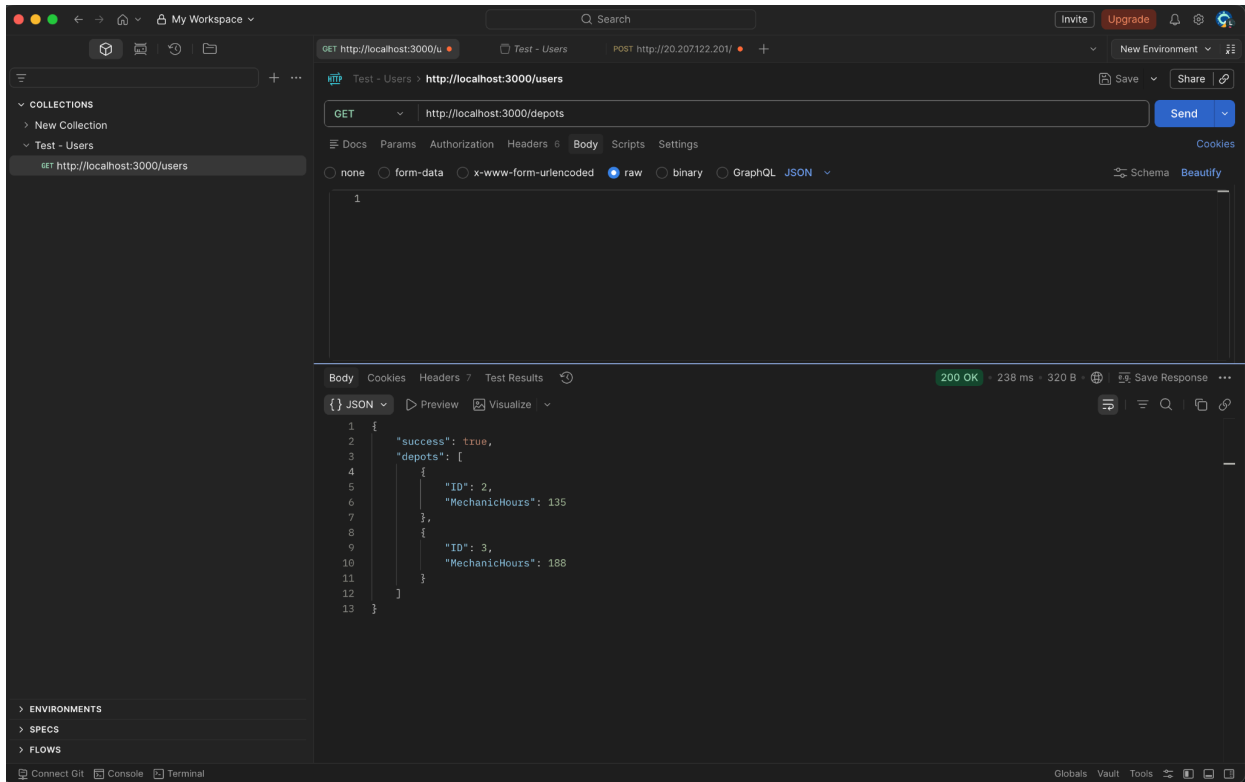
The system:

- Fetches depot and vehicle data
- Schedules maintenance tasks
- Allocates mechanic hour budgets
- Prioritizes tasks using scheduling logic
- Generates optimized schedules

## API Endpoints

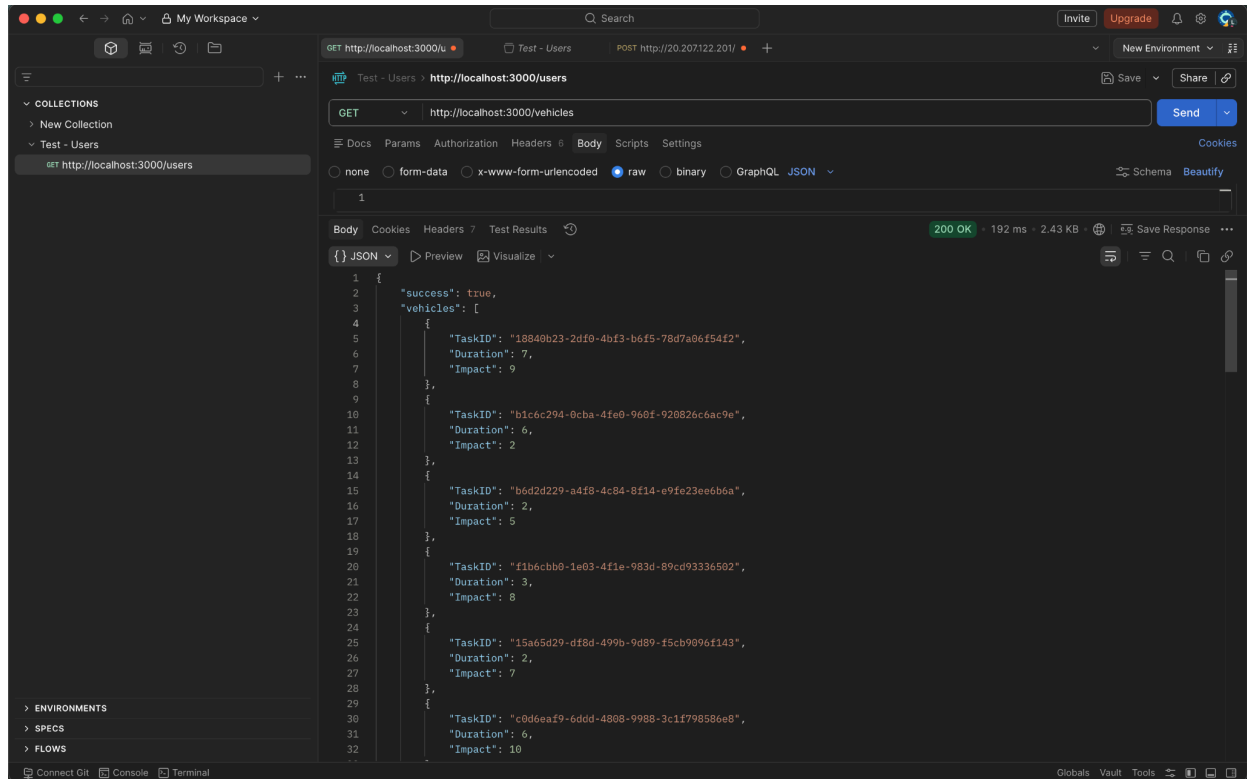
### Get Depots

GET /depots



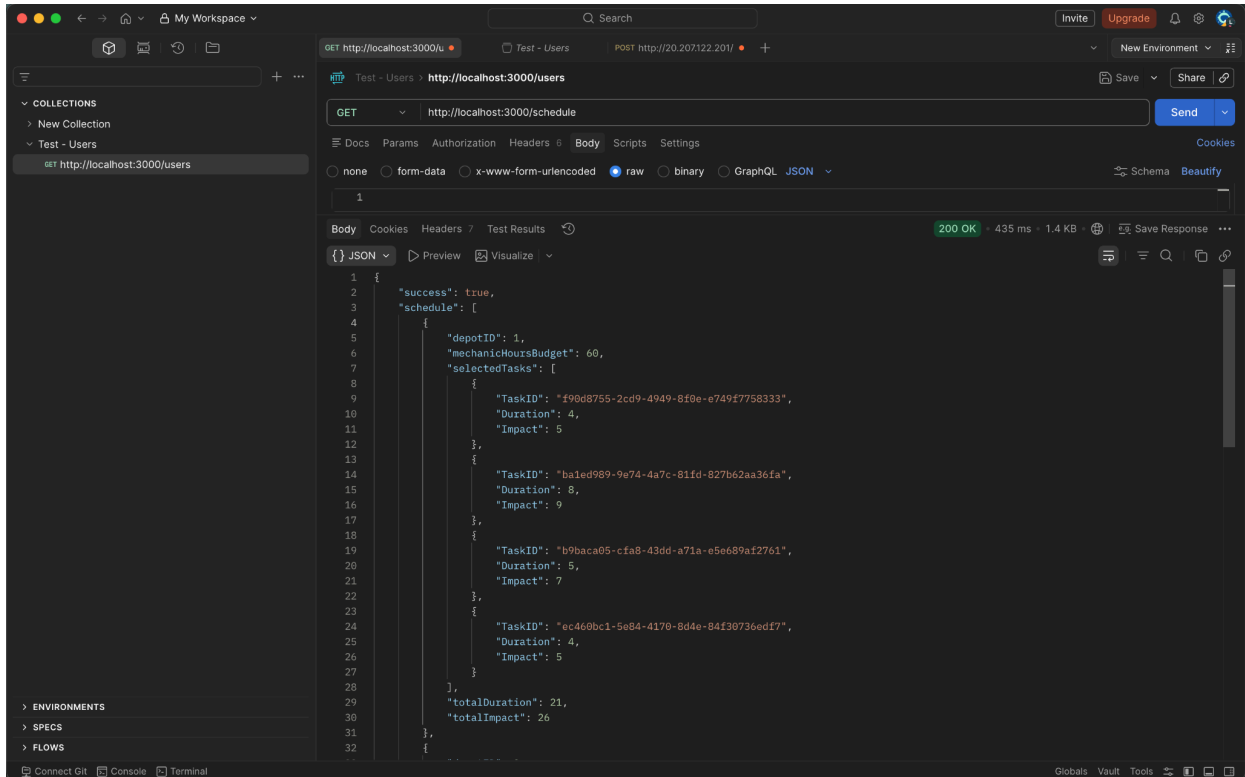
# Get Vehicles

GET /vehicles



# Generate Schedule

GET /schedule



## Example Schedule Response

```
{
  "success": true,
  "schedule": [
    {
      "depotID": 2,
      "mechanicHoursBudget": 135,
      "selectedTasks": [],
      "totalDuration": 0,
      "totalImpact": 0
    }
  ]
}
```

## 2. Notification System

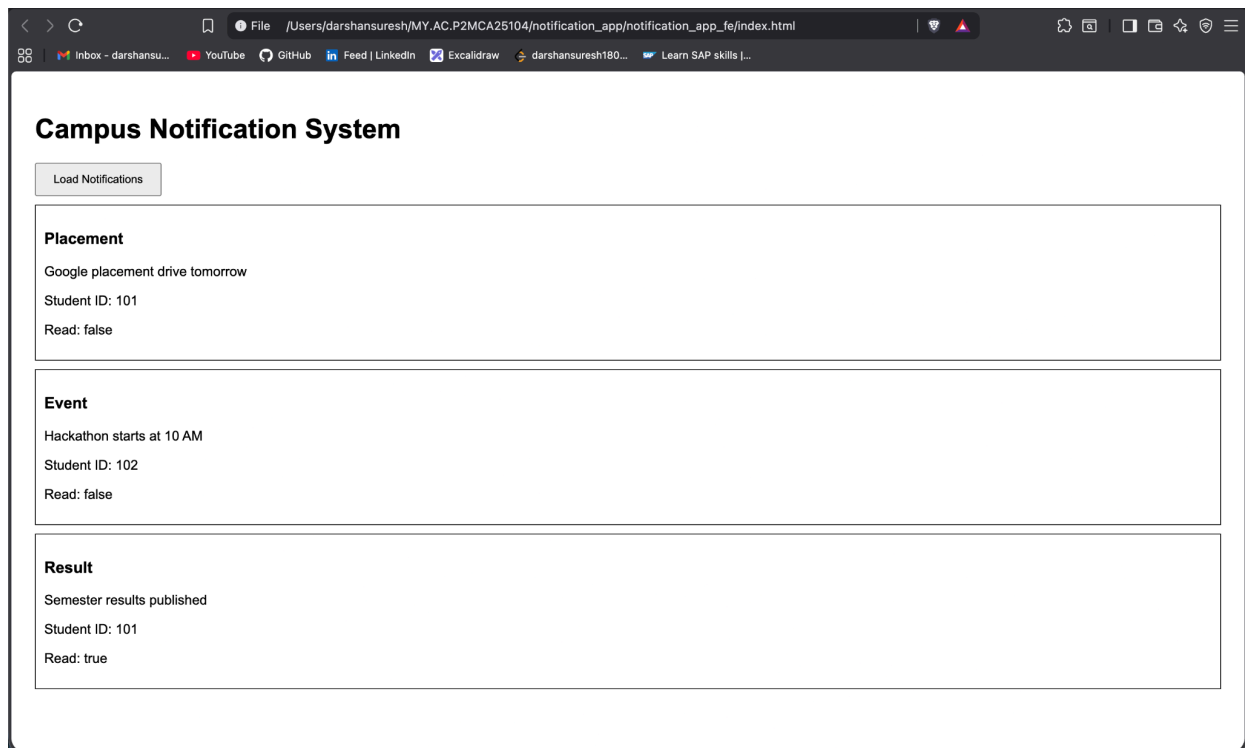
### Description

A campus notification system developed using:

- Express.js backend
- HTML/CSS/JavaScript frontend
- REST APIs
- Middleware authentication

The system supports:

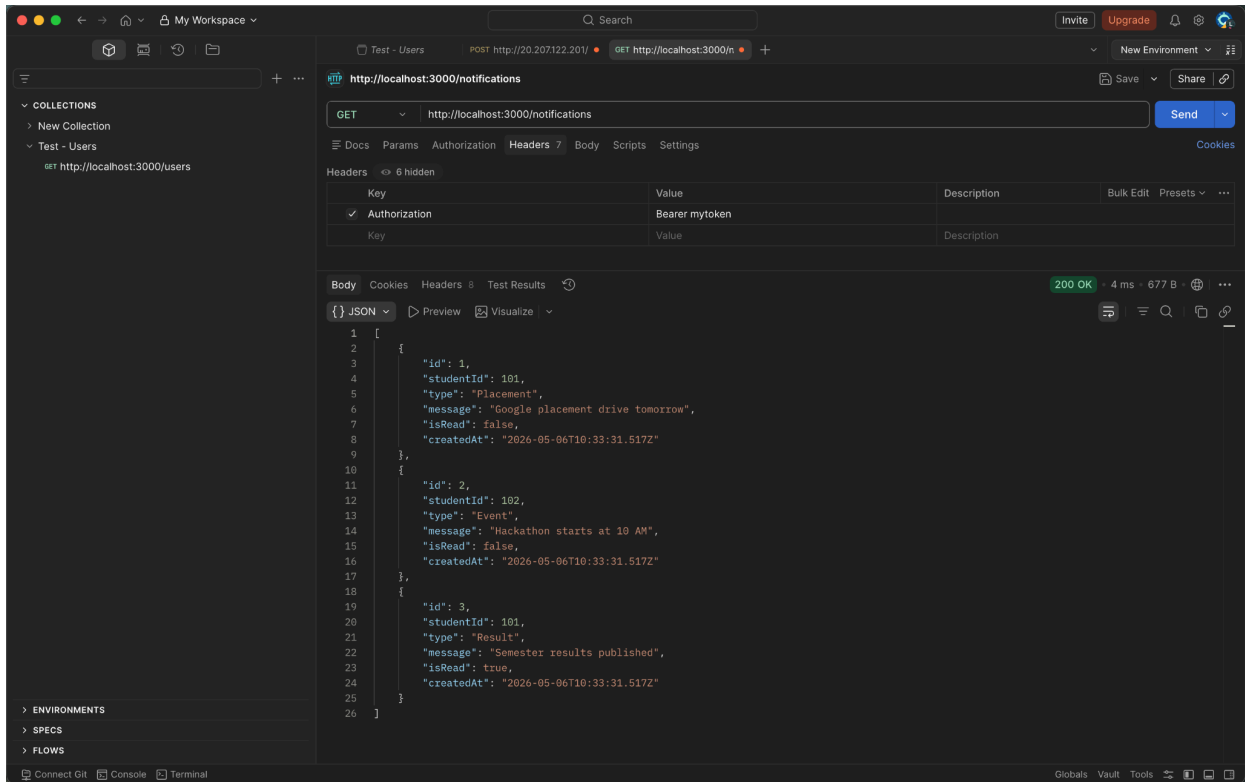
- Notification creation
- Read/unread tracking
- Notification deletion
- Frontend integration



# Notification APIs

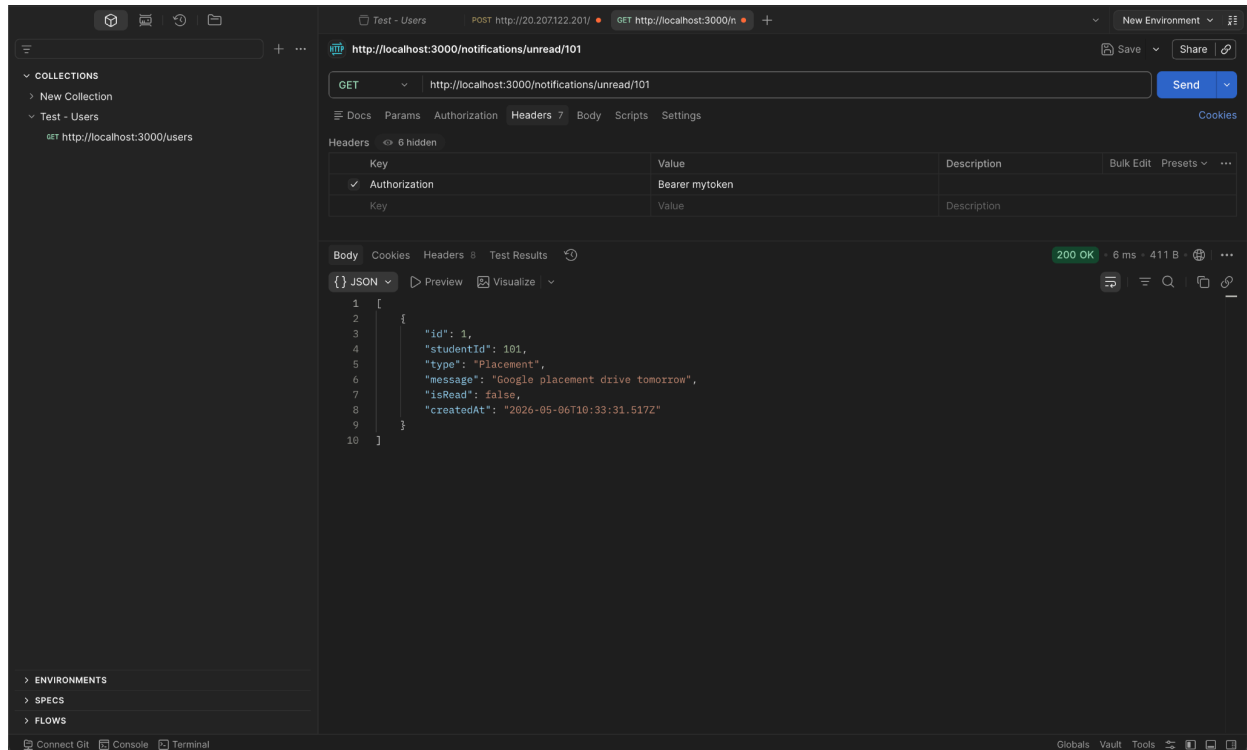
## Get All Notifications

GET /notifications



# Get Unread Notifications

GET /notifications/unread/:studentId

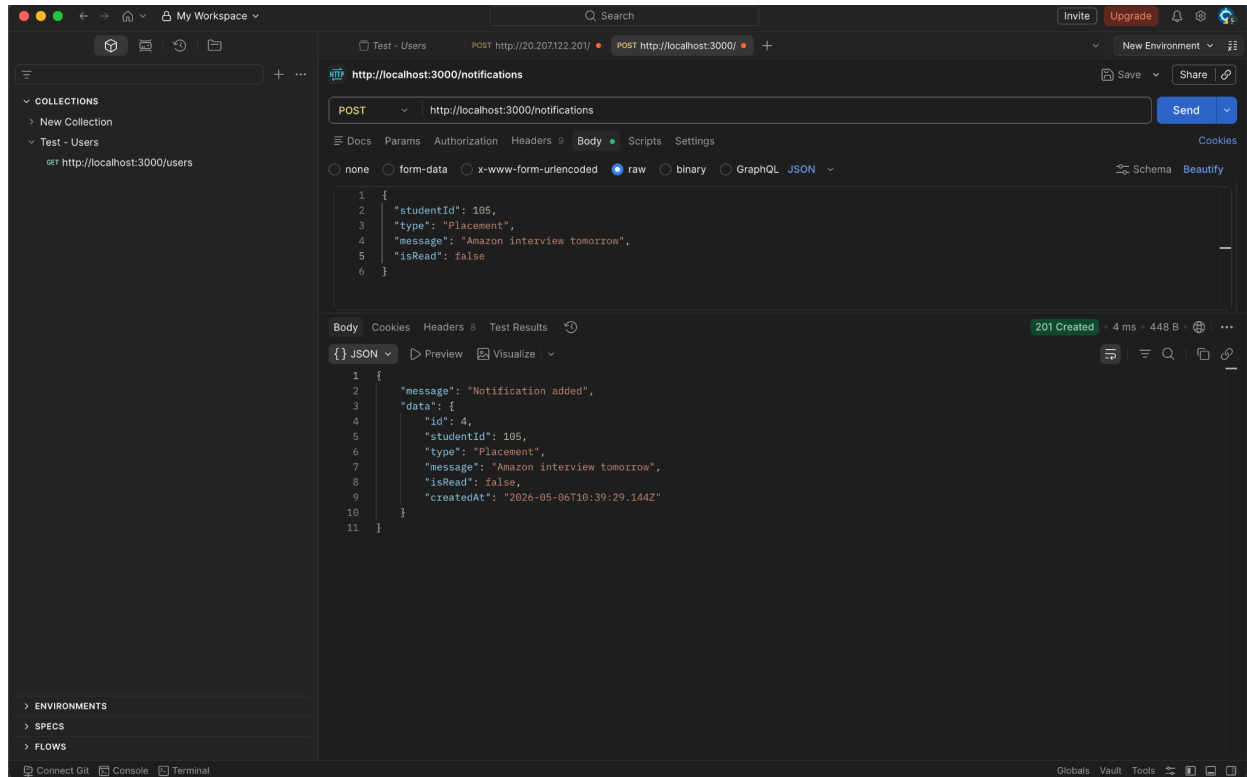


# Create Notification

POST /notifications

Example Body:

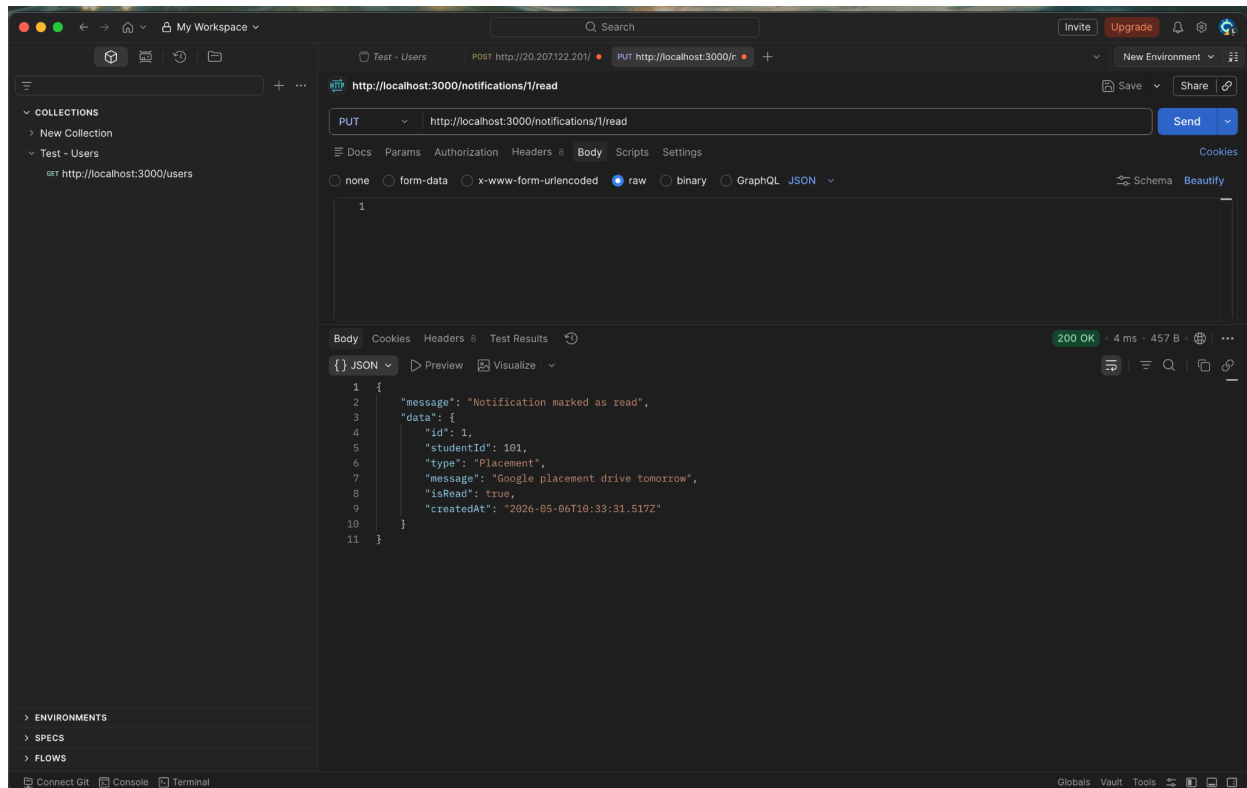
```
{
  "studentId": 105,
  "type": "Placement",
  "message": "Amazon interview tomorrow",
  "isRead": false
}
```



# Mark Notification as Read

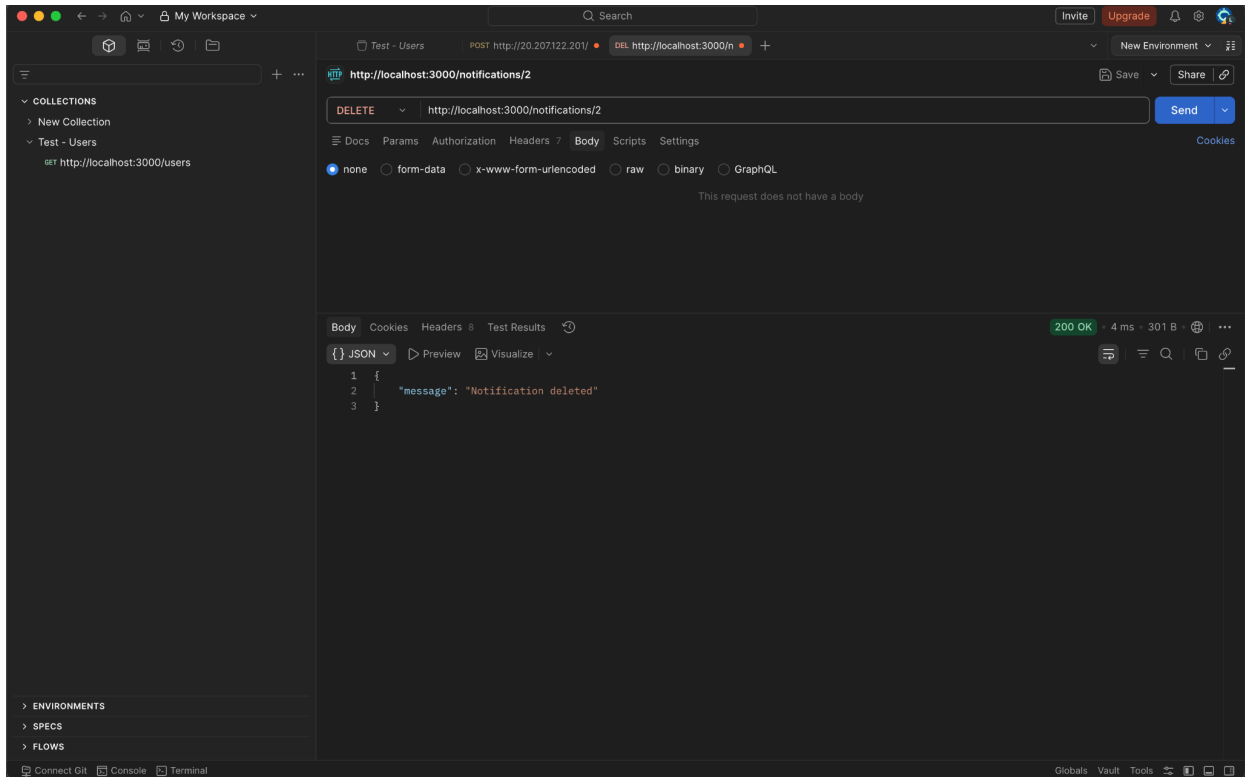
PUT /notifications/:id/read





## Delete Notification

DELETE /notifications/:id



## 3. Logging Middleware

### Description

Custom logging middleware developed using Express.js.

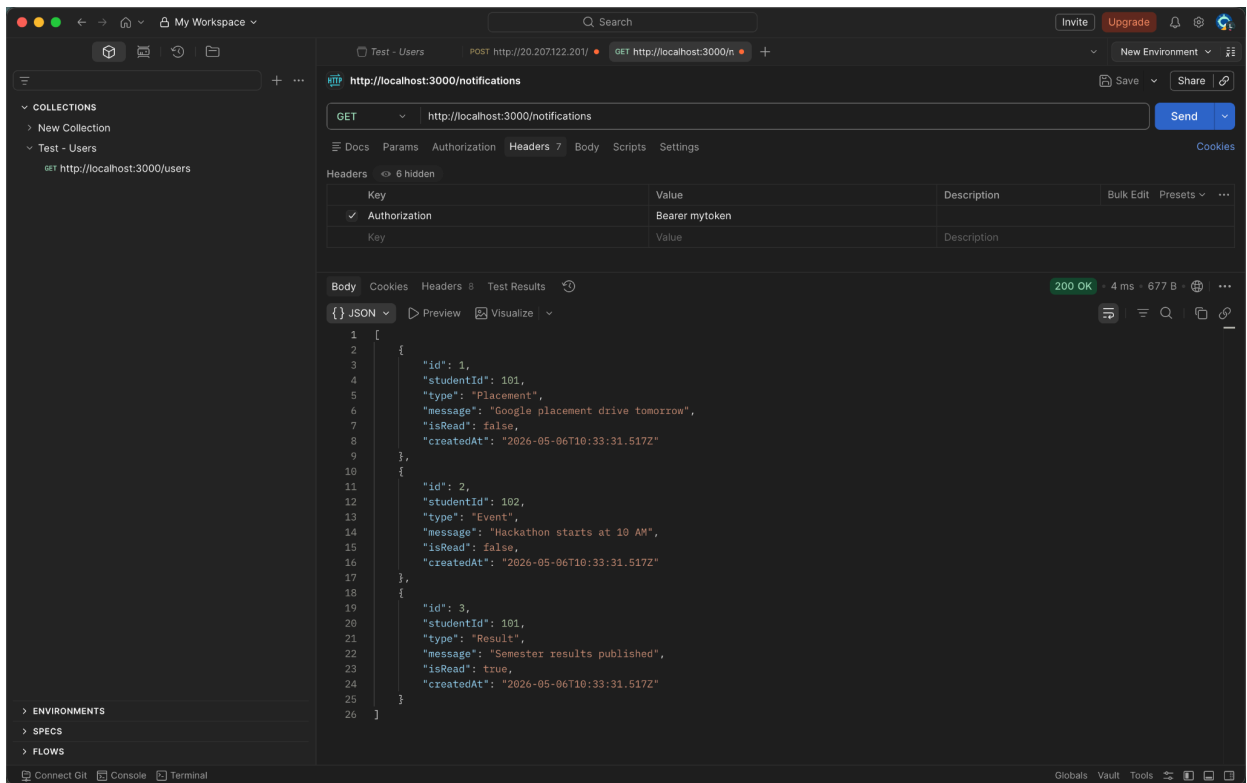
The middleware logs:

- API requests
- Request method
- URL
- Timestamp
- Warning logs
- Information logs

### Example Logs

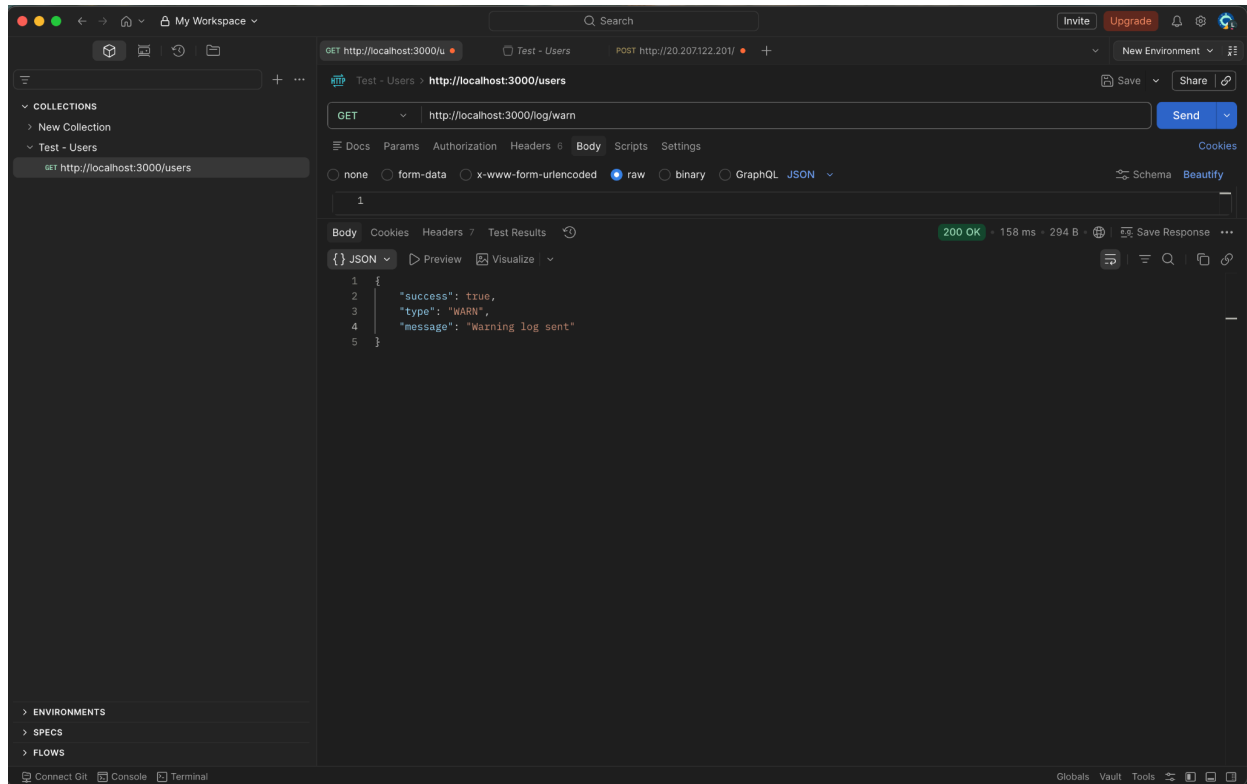
# Info Log

[INFO] GET /notifications



# Warning Log

[WARN] Unauthorized request detected



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